the identification of precipitates. What colour would the following precipitates give to a flame?

(a) $CaCO_{3(s)}$
(b) $PbCl_{2(s)}$
(c) $SrSO_{4(s)}$
(d) $Cu(OH)_{2(s)}$

Applying Inquiry Skills

4. Complete the **Analysis** and **Evaluation** sections of the following report.

Question

What ions are present in the solutions provided?

Experimental Design

The solution colour is noted and a flame test is conducted on each solution.

Evidence

Table 4: Solution and Flame Colours

Solution	Solution colour	Flame colour
A	colourless	violet
B	blue	green
C	colourless	yellow
D	colourless	yellow-red
E	colourless	bright red

Analysis

(a) Which ions are present in solutions A–E?

Evaluation

(b) Critique the Experimental Design.

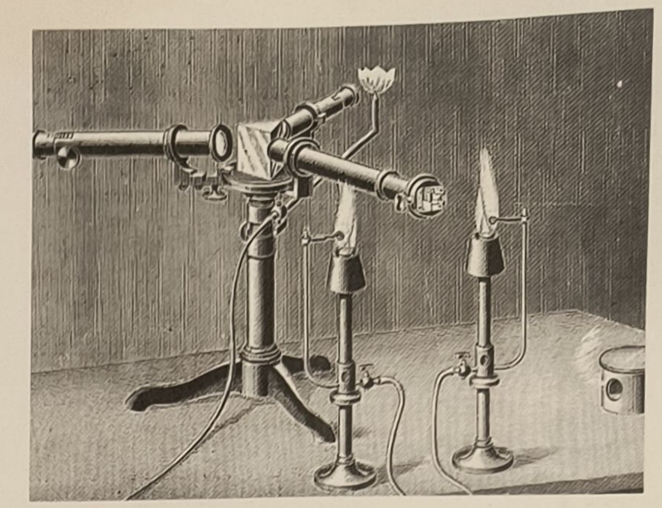

Figure 3
In 1860 Bunsen and Kirchhoff developed techniques of spectroscopy for analysis.

Figure 4
The spectrophotometer is a valuable tool, essential for precise qualitative and quantitative analyses in many areas of science.

Sequential Qualitative Chemical Analysis

In addition to colours of solutions and flames, analytical chemists have created very specific qualitative tests for ions in aqueous solution. These specific tests use the precipitation of low-solubility products. The chemist plans a double displacement reaction involving one unknown solution and one known solution, and predicts that, if a precipitate forms, then a certain ion must have been present in the unknown solution. Solubility tables help the chemist (and us) to choose reactants that will produce precipitates as evidence of the presence of specific ions. This type of test is called a **qualitative chemical analysis.**

qualitative chemical analysis: the identification of substances present in a sample; may involve several diagnostic tests

Suppose you were given a solution that might contain either lead(II) ions or strontium ions, or possibly both or neither. How could you determine which ions, if any, were present? You could perform a qualitative analysis. For this, you must design an experiment that involves two diagnostic tests, one for each ion—that is, tests that definitely identify a particular substance. Of course, it is hard to identify an ion in a solution unless it forms a compound with low solubility. In other words, unless it forms a precipitate. So you choose reactants that, if lead(II) ions or strontium ions were present, would form a precipitate.

Test for the Presence of Lead(II) Ions

Let's test for the possible presence of lead(II) ions in a solution, before going on to test for the presence of strontium ions (Figure 5, page 344). Chloride ions form a low-solubility compound with lead(II) ions, so if lead(II) ions are present

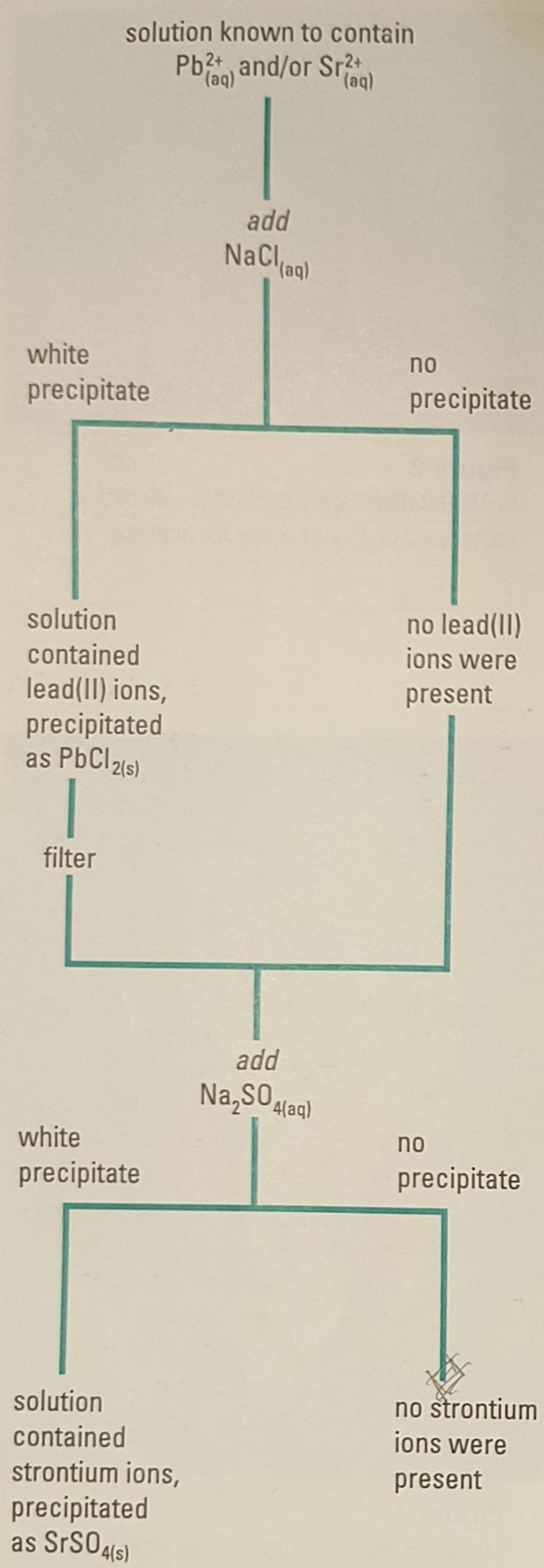

Figure 5
Reading down, we see that this is one experimental design for analyzing a solution for lead(II) and/or strontium ions. In this example, the two tests could not be done in reverse order, because both lead(II) and strontium ions precipitate with sulfate ions.

limiting reagent: a reactant that is completely consumed in a chemical reaction

excess reagent: the reactant that is present in more than the required amount for complete reaction

in a solution, a precipitate forms when chloride ions are added. An appropriate source of chloride ions would be a solution of sodium chloride—a high-solubility chloride compound. Refer to the solubility table (Appendix C) to confirm that strontium ions do not precipitate with chloride ions, but that lead(II) ions do.

If sodium chloride solution is added to a sample of the test solution and a precipitate forms, then lead(II) ions are likely present and the following reaction has taken place.

$$Pb^{2+}_{(aq)} + 2\,Cl^{-}_{(aq)} \rightarrow PbCl_{2(s)}$$

We add an excess of sodium chloride so that there are sufficient chloride ions to precipitate all of the lead(II) ions, leaving none in solution. This makes certain that there are no leftover lead(II) ions to interfere with a subsequent test for strontium ions.

When a sample under investigation (e.g., our unknown solution) is combined with an excess quantity of another reactant (e.g., sodium chloride), all of the sample reacts. The reactant that is completely used up (in this case, lead(II) ions) is called the **limiting reagent**. The reactant that is present in more than the required quantity (the chloride ions) is called the **excess reagent**.

Our precipitate indicates that lead(II) ions were present in the solution. Now we need to test our solution for strontium ions. First, however, we should remove the lead(II) chloride precipitate by filtering the solution. The remaining filtrate can then be tested for strontium ions.

Test for the Presence of Strontium Ions

What strontium-containing compound would form a precipitate? A look at the solubility table shows us that a compound containing sulfate ions and strontium ions is relatively insoluble. If a highly soluble sulfate solution (e.g., sodium sulfate) is added to the filtrate, and a precipitate forms, then strontium ions are likely to be present. (In the unlikely event that there are still some lead(II) ions in solution, they will react with the sulfate ions and would precipitate also.) If $Sr^{2+}_{(aq)}$ is present, then

$$Sr^{2+}_{(aq)} + SO^{2-}_{4(aq)} \rightarrow SrSO_{4(s)}$$

If we see a precipitate form, it is likely to be strontium sulfate. This indicates the presence of strontium ions in the original solution. We could also confirm the presence of strontium ions with a flame test.

In our example, both the unknown solution and the diagnostic-test solutions contained dissociated ions. If a precipitate forms, collisions must have occurred between two kinds of ions to form a low-solubility solid. The chloride and sodium ions present in the diagnostic-test solutions are spectator ions.

By carefully planning an experimental design for qualitative chemical analysis, you can do a sequence of diagnostic tests to detect many different ions, beginning with only one sample of the unknown solution.

We can use our experimental design as a guide for planning diagnostic tests for many ions (**Figure 6**, page 346).

SUMMARY

To complete a sequential analysis involving solubility, follow these steps. (These steps are written for cation (metal ion) analysis. For anion analysis, reverse the words cations and anions.)

1. Locate the possible cations on the solubility table.
2. Determine which anions precipitate the possible cations.
3. Plan a sequence of precipitation reactions that uses anions to precipitate a single cation at a time.
4. Use filtration between steps to remove cation precipitates that might interfere with subsequent additions of anions.
5. Draw a flow chart to assist your testing and communication.

Recognize that the absence of a precipitate is an indicator that the ion being tested for is not present. Realize that sometimes an experimental design can be created where parallel tests rather than sequential steps are used. For example, you can test for the presence or absence of calcium and mercury(I) ions by adding sulfate and chloride ions to separate samples of a solution.

Investigation 7.5.1

Sequential Chemical Analysis in Solution

A security guard noticed a trickle of clear liquid dripping from a drainage channel in the warehouse of a chemical plant. The guard alerted the chemists. To find out which of the containers of chemical solutions was leaking, the chemists needed to find out what ions were in the leaked solution. They decided to test for the most likely ions first.

In this investigation you are one of the chemists identifying the spill. Your purpose is to test a sample of the spilled solution for the presence of silver, barium, and zinc ions. You have to plan and carry out a qualitative chemical analysis using precipitation reactions, and analyze the evidence gathered.

Complete the **Materials**, **Experimental Design**, **Procedure**, and **Analysis** sections of the investigation report.

Question

Are there any silver, barium, and/or zinc ions present in the leaked solution?

Experimental Design

(a) Plan a sequential qualitative analysis involving precipitation and filtration at each step. Use sodium chloride, sodium sulfate, and sodium carbonate to test for silver, barium, and zinc ions. Draw a flow chart to communicate the experimental design and procedure.
(b) Suggest flame tests on the precipitates that would increase your confidence in the qualitative analysis.
(c) Create a list of Materials and write out your Procedure. Include any necessary safety precautions and special disposal arrangements.

Procedure

1. With your teacher's approval, carry out your investigation. Record your observations at every step.

Analysis

(d) Analyze your evidence to answer the Question.

INQUIRY SKILLS

- ○ Questioning
- ○ Hypothesizing
- ○ Predicting
- ● Planning
- ● Conducting
- ● Recording
- ● Analyzing
- ○ Evaluating
- ● Communicating

See Handling Chemicals Safely in Appendix B.

Barium solutions are toxic and should be treated with care. Avoid swallowing and avoid contact with the skin. Wear gloves.

Silver solutions are corrosive and must be kept away from the eyes and skin, and must not be swallowed. Wear eye protection and do not rub your eyes.

Zinc nitrate is harmful if swallowed.

Dispose of all waste substances in a special container labelled "Heavy Metal Waste."